MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE

Gwalior (M.P.), INDIA

Deemed University

(Declared under Distinct Category by Ministry of Education, Government of India)

A PROJECT REPORT

ON

AI-POWERED REAL-TIME BATTERY FITNESS ASSESSMENT PLATFORM

Bachelor of Technology

in

Centre for Artificial Intelligence

SUBMITTED BY

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Enrollment No: **BTAI25O1001**

Artificial Intelligence(AI) | Semester 3

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Artificial Intelligence(AI) | Semester 3

UNDER THE SUPERVISION OF

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CENTRE FOR ARTIFICIAL INTELLIGENCE

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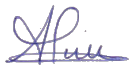
Academic Session: July-Dec 2026

DECLARATION

We hereby declare that the project work entitled **"AI-Powered Real-Time Battery Fitness Assessment Platform"** is an authentic record of our own research and engineering implementation carried out under the supervision of **Dr. Neelam Arya**, Assistant Professor, Centre for Artificial Intelligence, Madhav Institute of Technology & Science, Gwalior, during the academic session **July-Dec 2026**.

We further declare that this report represents our own original work and, to the best of our knowledge, it contains no materials previously published or written by another person, nor has it been submitted for the award of any other degree or diploma at any institution. We confirm that all sources of information used in this report have been duly acknowledged in accordance with accepted academic practice.

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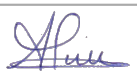
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SUPERVISOR:

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Date: 31 October 2026
Place: Gwalior, Madhya Pradesh

CERTIFICATE

This is to certify that the project report entitled

AI-POWERED REAL-TIME BATTERY FITNESS ASSESSMENT PLATFORM

has been carried out by the following students from the **Centre for Artificial Intelligence**, Madhav Institute of Technology & Science, Gwalior (M.P.), INDIA, in partial fulfillment of the requirements for the award of the Degree of **Bachelor of Technology** during the academic session **July-Dec 2026**:

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The above report has been examined and found to be satisfactory in scope, methodology, results, and presentation. It is hereby approved for submission in accordance with the ordinances of Madhav Institute of Technology & Science (Deemed to be University), Gwalior.

SUPERVISOR:

Date: 31 October 2026

Place: Gwalior, Madhya Pradesh

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TABLE OF CONTENTS

TITLE / CHAPTER	PAGE
Declaration	**ii**
Plagiarism Check Certificate	**iii**
Certificate	**iv**
Table of Contents	**v**
1. Project Overview	**1**
1.1 Abstract	•
1.2 Problem Statement	•
1.3 Objectives	•
1.4 Scope	•
2. Literature Review & Background	**2**
2.1 Background & Survey	•
2.2 Existing Systems & Limitations	•
2.3 Comparative Study & Proposed Work	•
3. Requirements Analysis & Specification	**3**
3.1 Functional Requirements	•
3.2 Non-Functional Requirements	•
3.3 Hardware & Software Prerequisites	•
4. System Design & Architecture	**4**
4.1 Architecture Description	•
4.2 System Modules & Workflow	•
4.3 Database & Data Flow Design	•
5. Technology Stack & Implementation	**5**
5.1 Technology Stack & Tools	•
5.2 Implementation Details & Modules	•
6. Testing & Results	**6**
6.1 Testing Methodology & Test Cases	•
6.2 Results & Outputs	•
6.3 Performance Analysis	•
7. Conclusion & Future Scope	**7**
7.1 Key Achievements	•
7.2 Limitations & Challenges	•
7.3 Future Scope	•
7.4 Conclusion	•
8. References	**8**
8.1 Citations & Bibliography	•
9. Annexure	**9**
9.1 Project Artifacts & URLs	•

CHAPTER 1

PROJECT OVERVIEW

1.1 Abstract

Built to bridge the gap between computer vision and everyday physical training, FitTrack is an automated web platform that evaluates workout movements from uploaded video footage. Instead of manual observation, the application harnesses YOLO26-Pose to extract skeletal keypoints and uses OpenCV to parse frames sequentially.

The core system features eight specialized evaluation scripts, each tuned to distinct movement mechanics to accurately count repetitions and measure performance. Django handles user accounts, secure authentication, and profile histories, while Celery paired with Redis manages asynchronous background queues to keep the web interface fast. As an operational prototype, the project successfully unifies video parsing, movement logic, and user management into a single platform, with future improvements focused on cross-environment validation.

1.2 Problem Statement

Traditional workout tracking depends a lot on people watching to count reps check how you move and keep track of progress. This manual process can make people tired. Often gives uneven results.

FitTrack solves this problem by using video to analyse movement. FitTrack finds bones and joints with pose estimation. Runs special checks for each exercise. FitTrack cuts out bias removes the need, for manual logs and gives instant fair performance feedback.

1.3 Objectives

- Create a place where people can use a web tool that works well on all devices to get automatic workout checks.
- Use YOLO26-Pose to get points of the human body.
- Use OpenCV to get frames from videos and change videos in a way.
- Write ways to check eight different kinds of exercise.
- Find out how times someone did an exercise and give a total score for the workout.
- Make sure each person has an account and can see their past sessions.
- Send tasks to special workers that run in the background using Celery and Redis.

1.4 Scope

FitTrack currently supports eight exercises with full end-to-end functionality. It handles user onboarding processes video asynchronously and maintains structured history logs. Built using Python libraries and a modular web framework the prototype is designed to be flexible and easy to expand with improvements.

System reliability depends on conditions, like camera angle, lighting and how tightly clothes fit on the body. Because of this even though the platform works well and offers a foundation the scores it produces need more real-world testing and validation. Until then these scores should not be considered fitness benchmarks.

CHAPTER 2

LITERATURE REVIEW & BACKGROUND

2.1 Background & Survey

Human pose estimation algorithms map body points—shoulders, wrists, hips ankles—from pictures and videos. Early methods like OpenPose used Part Affinity Fields to track people at once. Later systems such, as MediaPipe made it easier to find landmarks on a phone or tablet. Now newer versions have put pose estimation into the YOLO system.

In FitTrack YOLO26-Pose finds the coordinates of body points.. Those coordinates alone do not say whether a workout move was done right. Turning the coordinate stream into real workout info needs rule■based logic. OpenCV helps play and edit video while the back end keeps records and runs the app.

2.2 Existing Systems & Limitations

Many fitness apps let users log workouts, track activity or follow exercise routines.. Logging workout data and analyzing movements from videos are not the same thing.

Pose estimation tools can find body points, like elbows or knees.. Knowing what those points mean for a specific exercise needs extra thinking. For example a bent knee might mean a squat or a lunge depending on the movement.

Camera angle, lighting or wrong point detection can mess up the results. These problems make it hard to get feedback.

FitTrack tries to solve this by using pose detection and adding its exercise analysis. It looks at movement patterns. Compares them to correct forms. This way it helps users understand what they are doing right or wrong.

2.3 Comparative Study & Proposed Work

FitTrack was built by combining three parts: detecting body position in space analyzing movement based on rules and managing records in a structured way. I saw that using the method for everyone doesn't work when looking at how the body moves. So I created eight evaluation systems instead of using the same rules for all exercises. When these systems work with the history of each user they form a system, for tracking fitness automatically.

CHAPTER 3

REQUIREMENTS ANALYSIS & SPECIFICATION

3.1 Functional Requirements

- **User Management: Secure user registration, login authentication and personalized dashboard access.**
- **Skeletal Tracking: Detect body landmarks from exercise videos with YOLO26-Pose.**
- **Conditional Evaluation: Run analysis for each of the eight supported exercises.**
- **Metric Computation: Count repetitions, check form accuracy and score performance automatically.**
- **History Archiving: Save all sessions so users can look back at earlier assessments.**
- **Body Calibration: Let users enter their measurements and estimate body proportions using the detected keypoints.**
- **Asynchronous Queueing: heavy video parsing jobs, to Celery and Redis workers so the main app stays fast.**

3.2 Non-Functional Requirements

- Performance: I notice fast page loads and smooth UI because tasks run in the background.
- Security: I trust that strong login and permission checks keep my data safe.
- Reliability: I appreciate when errors are handled calmly if uploads break or file types are wrong.
- Usability: I like the simple tidy interface that lets me move around easily and see metrics clearly.
- Maintainability: I know the system is built with separate parts that let me add new workout routines without fuss.
- Scalability: I know the backend can handle more users, at once and heavier work without slowing down.

3.3 Hardware & Software Prerequisites

Hardware Requirements

FitTrack can run on a computer with an Intel or AMD multi-core processor and at least 8 GB of RAM recommended. Enough storage is needed for the project files and uploaded exercise videos. Users can provide videos through a webcam or use prerecorded videos. A GPU is optional but may help process videos faster.

Software Requirements

The project uses Windows or Linux as its operating system and Python as its main programming language. Django handles the backend, while YOLO26-Pose detects body keypoints and OpenCV processes video frames. Celery and Redis support background task processing. SQLite or PostgreSQL can be used to store application data. The frontend is built with HTML, CSS and JavaScript, and Docker supports container management. A modern web browser is required to access the application.

CHAPTER 4

SYSTEM DESIGN & ARCHITECTURE

4.1 Architecture Description

FitTrack has three main parts. The frontend is developed using HTML, JavaScript and Canvas to display the interface and assessment results. Django manages the backend, including user login, application functions and assessment records. Celery and Redis are used to handle selected tasks in the background, helping separate lengthy processing tasks from normal application requests.

4.2 System Modules & Workflow

When a user uploads an exercise video the application looks at each frame to find body points. The exercise module checks how these points move over time to figure out what happened. After the analysis is done the backend saves the result. Makes it available for the user to see. Every exercise has its set of rules, for how the movement is judged.

4.3 Database & Data Flow Design

A user information database keeps details such as body measurements as well as fitness evaluation data. Upon uploading a video, it becomes processed by the backend systems. The results obtained are then presented to users on their user interfaces. Access checks ensure that users receive the information they are entitled to receive.

CHAPTER 5

TECHNOLOGY STACK & IMPLEMENTATION

5.1 Technology Stack & Tools

FitTrack uses the following technologies for its development:

- Python: the programming language used by FitTrack.
- Django: a backend framework that handles requests, authentication and application logic for FitTrack.
- YOLO26-Pose: a tool that detects human body keypoints during exercise analysis for FitTrack.
- OpenCV: a library that processes uploaded videos and extracts frames for FitTrack.
- Celery and Redis: services that manage background video■processing tasks for FitTrack.
- SQLite/PostgreSQL: databases that store user information and assessment records for FitTrack.
- HTML, CSS and JavaScript: technologies that develop the frontend interface for FitTrack.
- Docker: a tool that supports application containerisation, for FitTrack.

5.2 Implementation Details & Modules

The platform has modules, for authentication, video uploading, pose estimation, exercise analysis and result management. Uploaded videos are processed to detect body keypoints and those body keypoints are analysed using logic. The generated assessment results are stored in the database. Displayed on the dashboard.

CHAPTER 6
TESTING & RESULTS

6.1 Testing Methodology & Test Cases

Testing was done to see if the platform works well if it is dependable and if the way people get access is correct. The main testing tasks were:

Functional Testing: Made sure that logging in uploading videos, analyzing exercises and creating results all worked.

Module Testing: Looked at each part separately like how the body position's detected and how exercises are analyzed.

Access Control Testing: Made sure that people only saw the data they were allowed to see.

Database Testing: Checked that user information and test results were stored and taken out properly.

6.2 Results & Outputs

Results

The project testing included fourteen automated tests and seventeen access-control tests. All seventeen access-control tests passed in the scenarios that were tested. These tests helped verify that the application behaves as expected and that user-data access restrictions work correctly.

Output

The system detects body keypoints from exercise videos analyzes exercise movements and generates assessment results. Users can view their exercise records and previous assessment details on the dashboard.

6.3 Performance Analysis

FitTrack takes the exercise videos that users upload and uses YOLO26-Pose and OpenCV to find body parts and look at how the movementsre done. A background task system, with Celery and Redis makes sure that analyzing the video does not stop things from working normally. How well the system works depends on how good the video's what kind of computer is used for processing and how long the video is. More improvements could make the processing faster. The analysis more accurate.

CHAPTER 7

CONCLUSION & FUTURE SCOPE

7.1 Key Achievements

FitTrack shows how well deep■learning pose estimation can work with fitness tracking. FitTrack's key achievements are: FitTrack accurately finds joint positions with YOLO26■Pose. FitTrack handles frame changes and video parsing smoothly with OpenCV. FitTrack uses custom movement rules to count repetitions, in eight exercises. FitTrack keeps data with secure login and stores personal records. FitTrack runs tasks in the background using Celery and Redis. FitTrack offers a dashboard where users can see their workout history.

7.2 Limitations & Challenges

Detection accuracy can fluctuate under poor ambient lighting or skewed camera angles.

Occlusion or blocked body parts can occasionally miscount repetitions.

Scoring accuracy relies on fixed mathematical thresholds that may need calibration for different body types.

Processing duration scales with video length and available hardware acceleration.

7.3 Future Scope

- Improving how accurately we track body positions and expanding the range of exercises
- Setting up scoring standards that work well across different age groups and backgrounds.
- Speeding up video analysis by using GPU clusters.
- Adding tools to compare progress over time and visual reports that show trends clearly.
- Creating apps to go with the system and making sure it runs smoothly in the cloud.

7.4 Conclusion

Developing FitTrack provided hands-on experience in merging computer vision pipelines with full-stack web engineering. The project establishes a solid technical prototype for automated workout evaluation. While the current system successfully fulfills its core objectives, future refinements in algorithmic thresholding and cross-environment testing will make its performance evaluations even more robust and dependable.

CHAPTER 8

REFERENCES

8.1 Citations & Bibliography

The following references were used for understanding and developing the technologies and concepts applied in FitTrack:

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CHAPTER 9

ANNEXURE

9.1 Project Artifacts & URLs

The following digital deliverables, code repositories, hosted environments, and resources are submitted for evaluation:

Artifact Title	URL / Repository Link
GitHub/ Source Code repository	https://github.com/AI-Engineer000/FitTrack-Platform
Live hosted application/ prototype	Not available
Demonstration video/ walkthrough	To be added
Dataset/ made	*Not provided*